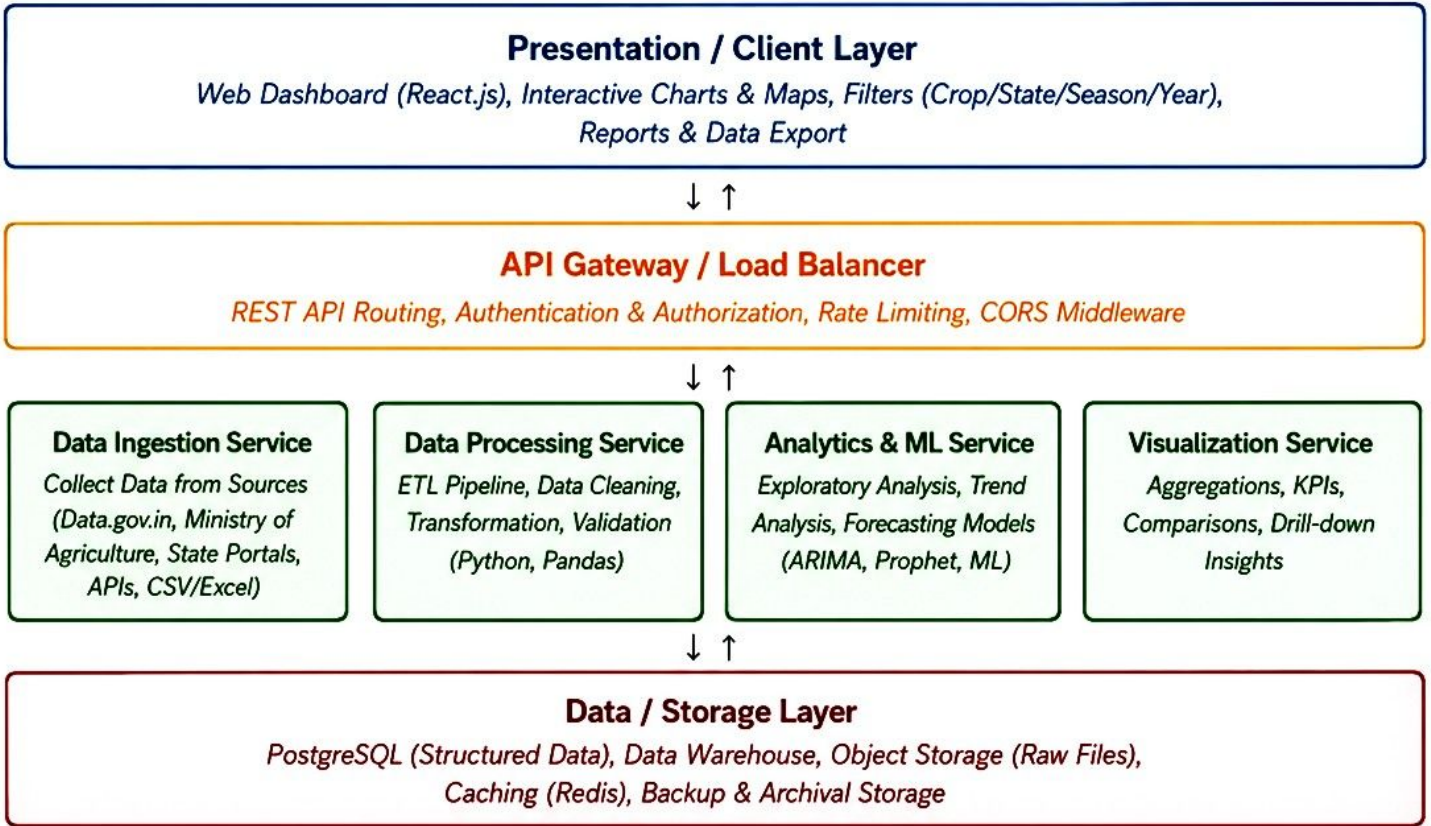


SOLUTION ARCHITECTURE

Date	3 July 2026
Team ID	
Project Name	India's Agriculture Crop Production Analysis
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



INDIA'S AGRICULTURE CROP PRODUCTION ANALYSIS

Empowering Farmers. Enriching Agriculture. Ensuring Food Security.

SKILL WALLET

A SmartBridge Product

COMPONENT DESCRIPTION TABLE

Component Name	Description / Role in Architecture	Technologies Used
 Presentation Layer	Provides user interfaces for farmers, agri-advisors, and administrators. Displays dashboards, crop production analytics, maps, reports, and alerts. Supports user registration, input forms, and real-time visualization of crop trends and forecasts.	React.js, Chart.js, Leaflet.js (GIS Maps), TailwindCSS, Axios
 API Gateway	Acts as the single entry point for all client requests. Routes requests to appropriate services, performs authentication and authorization, applies rate limiting, handles CORS, and ensures secure and efficient communication.	Kong / NGINX, FastAPI Gateway, JWT, OAuth2, Rate Limiting
 Microservices	Handles core business logic including data ingestion, processing, analytics, machine learning models (yield prediction, trend analysis, crop recommendation), and alert generation. Services are modular and independently scalable.	Python (FastAPI), Pandas, Scikit-learn, TensorFlow, Prophet, Redis
 Database / Data Layer	Stores structured and unstructured agricultural data including user data, crop statistics, soil & weather data, satellite imagery metadata, and model outputs. Supports transactional operations and analytics workloads.	PostgreSQL (Relational DB), MongoDB (NoSQL), AWS S3 / GCP Cloud Storage (Object Storage), Redis